Paper A v0.7 — The Physical (SU(3)) Yang–Mills Mass Gap via D_IV^5 Spectral Geometry. Companion to Paper B. v0.7 = v0.6 corrected by K1709: the mass gap is DEEPER than a single integer. On the REAL boundary $(S^4 \times S^1)/\mathbb{Z}_2$ the $\mathbb{Z}_2$ admits only k+m even, so the bare-S^4 'λ=4' mode (k=1,m=0, odd) is PROJECTED OUT — the surviving lowest modes are R-dependent ($4/R^2$, $4+1/R^2$). Energy-vs-Casimir resolved (Grace ⊕ Elie, §570): the ν-independent 4 was the SPATIAL S^4 vector eigenvalue; the ν/Δ piece is TIME energy, not mass. So the floor rolls back from '≥4' to EXISTENCE, and the honest value has THREE geometry-fixed opens: (i) the conformal weight Δ of the lowest gauge excitation, (ii) which surviving $\mathbb{Z}_2$-even mode is lowest, (iii) the radius R (= the ruler; decides $4/R^2$ vs $4+1/R^2$). The old '6' = conformal reading = the identity C_2 := n_C+1 (a time weight, not a mass); '7' = genus. The glueball 1720@0.6% is a SEPARATE valid prediction (built on C_2^int, not λ_1) — kept, but out of the room as gap-evidence. Existence-floor rock-solid; value = genuine open physics, no free dimensionless parameter beyond the ruler. Package intro HELD. NOT a derived value. Count HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral casc

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_IV^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.7 — K1709: on the real boundary $(S^4 \times S^1)/\mathbb{Z}_2$ the $\mathbb{Z}_2$ (k+m even) projects out the bare-S^4 "4", so the floor is EXISTENCE, and the honest value has three geometry-fixed opens — the conformal weight Δ, the $\mathbb{Z}_2$ coupling, and the ruler R. "6" was the conformal reading = the identity C_2 = n_C+1; the glueball (on C_2^int) is a separate valid prediction, out of the room as gap-evidence; package intro HELD.

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is the spectral gap of the Bergman–Laplacian on the bounded domain D_IV^5 — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. $\lambda_1 = C_2 = 6$ on our domain is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don't assert a shared eigenvalue).
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object); we do not lend the proton's four-decimal precision to the Yang–Mills gap (K1705). And the glueball is *out of the room* for the gap-*value* pin (K1708/K1709): 1720 = (11/6)·938 is built on C_2^int (the labeled integer primary), not on λ_1 (the boundary Laplacian gap), and its 11/6 carries C_2 = 6 — so it is a real, separate forward prediction but not independent evidence for the gap value. The value is pinned from the geometry alone (the GATE-KOIDE discipline that keeps Q

= 2/3 out of the Koide pin).

- Guard 3 (the symbol "C_2" itself — K1707). Three distinct objects all read "C_2" and all equal 6 at the physical point n = n_C = 5, so writing "the gap = C_2 = 6" hides *which one* is meant:
  - C_2^int := n_C + 1 = 6 — the *labeled integer primary* (one of the five integers). A definition.
  - C_2^flow(n) = 2n − 4 — the *commitment-flow Casimir* (a function of n; = 6 at n=5 only by the coincidence-at-n_C=5).
  - C_2^Berg = n + 1 — the *Bergman-representation Casimir* (a function; = 6 at n=5, and equal to C_2^int *identically*, since C_2^int is *defined* as n+1).

  The consequence (Cal, K1707): "$\lambda_1 = C_2 = 6$" as previously written is C_2^int = C_2^Berg — the same definition on both sides, an identity that *cannot fail* (P^2=P), not a derived tower-gap. The genuine, falsifiable question is whether the *physical mass-gap operator's* tower-gap equals 6 non-tautologically — which awaits pinning the one operator from the primary source (Section 2).
- The one residual (flat space). The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography → Osterwalder–Schrader) places on the flat 4D boundary literally interacting SU(3) Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

With those stated, the paper's claim is explanatory, not triumphal: BST reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain. The gap is nonzero because the domain is bounded (real physics, Section 2), with no free dimensionless parameter beyond the ruler R. Its *value* is genuine open physics: the bare-S^4 "4" is projected out by the $\mathbb{Z}_2$ on the physical boundary $(S^4{\times}S^1)/\mathbb{Z}_2$ (k+m even, K1709), and the honest value awaits three geometry-fixed opens — the conformal weight Δ, the $\mathbb{Z}_2$ coupling, and the radius R. The old "6" was the conformal reading = the *identity* C_2 = n_C+1 (a time weight, not a mass); "7" is the genus. The banked floor is existence: the gap exists, geometry-fixed, no dimensionless knob. The flat-$\mathbb{R}^4$ construction is carried as the large named open. Nothing here is "we solved Yang–Mills," and — after K1709 — nothing here claims a derived integer value; the value is honest open physics, not a stalled identity.

Abstract

We study the Yang–Mills mass gap for the physical gauge group SU(3) — color — derived (not assumed) from the substrate D_IV^5 in the companion Paper B. The mass-gap operator is a Laplacian on the physical Shilov boundary ∂_S D_IV^5 = $(S^4{\times}S^1)/\mathbb{Z}_2$, and its gap is nonzero because the domain is bounded — the spectrum is discrete and bounded below, and *compactness is the gap mechanism* (decompactify and the gap slides to zero: $\lambda_1(R) \to 0$). This existence is rock-solid, with no free dimensionless parameter beyond the ruler R (a Planck-unit, the theory's single dimensionful input). The *value*, however, is genuine open physics. A bare-S^4 computation gives a first eigenvalue 4 (tower k(k+3), degeneracies 1,5,14,30,55, first-level degeneracy n_C = 5), but the physical boundary is $(S^4{\times}S^1)/\mathbb{Z}_2$, whose

$\mathbb{Z}_2$ admits only k + m even and therefore *projects out* the $\lambda = 4$ mode (k=1, m=0, odd), leaving R-dependent survivors (K1709, Cal). The earlier "$\lambda_1 = C_2 = 6$" was the *conformal (SO(2)/time-circle) reading* — the bare 4 plus a +2 shift — coinciding with C_2 := n_C+1 = the Bergman-representation Casimir (an identity that cannot fail, Cal K1707), i.e. a *time* conformal weight, not a mass. So the honest value awaits three geometry-fixed opens (K1709): (i) the conformal weight Δ of the lowest gauge excitation, (ii) which surviving $\mathbb{Z}_2$-even mode is lowest, (iii) the radius R (the ruler, which decides $4/R^2$ vs $4+1/R^2$). The banked floor is existence itself: the gap exists, is geometry-fixed with no free dimensionless parameter beyond the ruler; its integer value is real open physics, not a stalled identity. *(The glueball 1720 MeV = (11/6)·938 is a separate valid prediction built on C_2^int, not λ_1 — kept, but out of the room as gap-evidence, K1708.)* The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_IV^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain's 4D conformal boundary (SO(4,2) ⊂ SO(5,2)), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* SU(3) YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^PC channels follow a linear conformal-energy ladder off one verified number (the genus of D_IV^5), with one blind leg, 2^{++} = g/n_C, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^PC tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat-$\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not "all G"

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: G = SU(3), color. The companion Paper B proves D_IV^5 is the unique substrate and that SU(3) is derived, not chosen. So "all G" is not BST's claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap: existence SOLID, value = three geometry-fixed opens on the real boundary $(S^4{\times}S^1)/\mathbb{Z}_2$ (K1709)

What is SOLID (real physics). The mass-gap operator is a Laplacian on the Shilov boundary of D_IV^5 — a positive operator whose spectrum *is* the arrow of time (the heat-semigroup generator). The domain is bounded; the state space H^2(D_IV^5) is square-integrable against the invariant Bergman measure, with the reproducing-kernel (Cauchy–Szegő / Shilov-boundary) structure fixing boundary behavior, so the spectrum is discrete and bounded below. Hence the gap is nonzero, and it is nonzero *because the domain is bounded*: decompactification ($R \to \infty$) sends $\lambda_1(R) \to 0$, killing the gap. *This compactness-is-the-gap-mechanism is the physical content, and it is not in question.* The gap has no free dimensionless parameter beyond the ruler — the geometry fixes everything up to the one dimensionful scale R (a Planck-unit, the same single input GR takes as G and QM as $\hbar$).

The bare-S^4 number "4" is real but is *projected out* on the physical boundary (Cal, K1709). On the *bare* four-sphere the Laplacian tower is λ_k = k(k+3) = {0, 4, 10, 18, …} with degeneracies 1, 5, 14, 30, 55 — a genuine S^4 spectrum (not S^6; there is no S^6 in BST), whose first level (k=1, λ=4) has degeneracy exactly n_C = 5. But the physical boundary is not bare S^4 — it is $(S^4 \times S^1)/\mathbb{Z}_2$, and the $\mathbb{Z}_2$ identification couples the spatial S^4 level k to the time-circle mode m and admits only k + m even. The $\lambda = 4$ mode is (k=1, m=0), for which k + m = 1 is odd — so it is projected out. The surviving lowest $\mathbb{Z}_2$-even modes are R-dependent (e.g. $4/R^2$ or $4 + 1/R^2$, depending on which even (k,m) is lowest). *So "$\lambda_1 = 4$" is not the physical gap; the bare-S^4 computation was a valid intermediate that the real boundary removes.*

Energy vs. Casimir, resolved (Grace ⊕ Elie, §570). The ν-independent number 4 is the spatial S^4 vector eigenvalue; the ν/Δ piece is time energy, not mass (the SO(2)/time-circle conformal weight). These are *different objects on different circles* — the spatial Laplacian and the time-conformal weight must not be summed into one integer as if they were a single Casimir.

The honest value: three geometry-fixed opens (K1709). The physical gap on $(S^4 \times S^1)/\mathbb{Z}_2$ awaits three quantities, each fixed by the geometry (no dimensionless knob), none yet pinned: 1. the conformal weight Δ of the lowest gauge excitation (the time-circle / SO(2) contribution); 2. the $\mathbb{Z}_2$ coupling — which surviving k+m-even mode is the lowest; 3. the radius R (= the ruler) — which decides whether $4/R^2$ or $4 + 1/R^2$ wins.

These are genuine open physics, not a stalled identity. *(Use Heat_Trace's own three-scale language: λ_1, $c_2 = 11$, and $C_2 = 6$ are each a distinct object — do not collapse them.)*

Why the old "6" and "7" were never the gap. "6" was the conformal reading — the bare 4 plus a +2 shift — which coincides with C_2 = n_C+1 = the Bergman-rep Casimir (the identity Cal flagged, K1707; Guard 3), i.e. the conformal weight equated with its own definition, not an independent gap. "7" is the genus g = n_C+2, a different invariant. Neither is λ_1 on the real boundary.

reading	value	status
bare-S^4 Laplacian first level, k(k+3)	4	real, but $\mathbb{Z}_2$-projected-out — not the physical gap (K1709)

reading	value	status
conformal reading (4 + ρ-shift 2)	6	= C_2 = n_C+1 — the identity (a time weight, not a mass)
Bergman-rep Casimir, n+1	6	≡ the identity (same definition)
genus / embedding invariant, g	7	a different invariant, not the gap
physical gap on $(S^4{\times}S^1)/\mathbb{Z}_2$	awaits Δ + $\mathbb{Z}_2$-coupling + R	genuine open (K1709)

The glueball is a separate valid prediction, but out of the room for the gap (K1708/K1709). The glueball 1720 MeV = (11/6)·938 is a real forward prediction (Section 7) built on C_2^int (the labeled integer primary), not on λ_1 (the boundary Laplacian gap). It stays as a prediction, but it cannot corroborate the gap value — the same GATE-KOIDE discipline that keeps Q = 2/3 out of the Koide pin. The gap is pinned from the geometry alone.

Tier: gap EXISTENCE = SOLID (bounded ⇒ gap, real physics; no free dimensionless parameter beyond the ruler). Gap VALUE = genuine open — three geometry-fixed unknowns (Δ, the $\mathbb{Z}_2$ coupling, R), the bare-S^4 "4" projected out. Floor = EXISTENCE (not ≥ 4). *Guard 1 holds: a gap of a boundary Laplacian on D_IV^5, not asserted to be the $\mathbb{R}^4$ YM Hamiltonian's gap.*

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — "you treat only SU(3)" — is answered by showing G is not a free input. Paper B establishes D_IV^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing dim_C = 5 and N_c = 3 (the dimension-free output of the proved T1829). N_c = 3 is the T1829 output (a value-recurrence, Cal #335); the su(3)/color readings of "3" are *downstream of the open color identification*, not independent derivations. So SU(3) is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix rank = 2 (N_c = rank^2−1), and rank generates {n_C, C_2, g} (T2491). The "all G" wall is dissolved.

4. W1 folded: the D_IV^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_IV^5 spectral theory carries it — on the domain's 4D conformal boundary, which exists because D_IV^5 carries the 4D conformal group SO(4,2) ⊂ SO(5,2) (the same edge where light and electromagnetism live):

OS datum	D_IV^5 realization	tier
rigorous Hilbert space	the Hardy space H^2(D_IV^5)	SOLID
Hamiltonian with a gap	Laplacian on the real boundary $(S^4 \times S^1)/\mathbb{Z}_2$; gap nonzero (bounded ⇒ gap), no dimensionless knob beyond R; value awaits Δ + $\mathbb{Z}_2$-coupling + R (bare-S^4 "4" $\mathbb{Z}_2$-projected-out, K1709)	SOLID (existence); value open
reflection positivity	tube-type (Cayley → $T(\Omega)$); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta > 0$ (nonzero, geometry-fixed)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is SO(5,2)-equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* SO(5,2) positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. *(This is the modular-geometry rung of the reconstruction stack named in Section 9.)*

6. The one residual: interacting Yang–Mills, or generalized-free? (= "is the Rehren edge literally YM?")

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting SU(3) Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3's naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the residual is to show the reconstructed theory is genuinely *interacting*. - F1056's naming (Rehren-edge = literally-YM): Rehren's algebraic holography is a rigorous bulk↔boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* SU(3) YM, not a cousin.

These are the same open piece. "Genuinely interacting" and "literally the physical YM net" are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. "Non-commutative ==> interacting" is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n-point functions ($n \geq 3$). The YM self-interaction is the vertex [A,A], nonzero precisely because the gauge algebra is non-abelian. So

interacting <=> the gauge algebra is genuinely non-abelian su(3) <=> the cross-channel glueball spectrum matches SU(3) (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian su(3) (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^PC tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal's trim of the v0.2 over-claim "derived / landed interacting").

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D\_IV^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible J^PC components, each of which must have eigenstates for the substrate's operator algebra to close:

channel	operator	what it commits
0^{++}	$\mathrm{Tr}(F^2)$	energy density (the action)
0^{-+}	$\mathrm{Tr}(F\,\tilde{F})$	topology (Pontryagin density)
2^{++}	Tr(F^$\mu\rho$ F^ν_ρ) traceless	curvature (the stress tensor)
1^{+-}	Tr(F[D,F])	derivative structure

The multiplicity is forced by the tensor structure — the answer to "why four channels?" Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The oddballs (0^{+-}, 1^{-+}, 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the SO(2) dilatation) on the holomorphic discrete series on H^2(D_IV^5), diagonal in the K-type basis by Schur:

m ∝ E = λ_0 + (energy step), λ_0 = genus(D_IV^5) = n_C = 5 (verified from the multiplicity formula p = (r−1)a + b + 2 = 5).

Energy step = SO(5) harmonic degree = spin J; the parity-odd (Hodge-dual) sector carries the half-canonical twist n_C/2. With one dimensionful anchor (seat = π^5·m_e = 156.4 MeV; m(0^{++}) = c_2·seat = 1720 MeV):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	J = 2	7/5	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	n_C/2	3/2	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	1 + n_C/2	17/10	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the 2^{++} = g/n_C leg is fully blind (genus + spin only, nothing read from the data, lands because g = n_C + rank). The 0^{-+} and 1^{+-} use the half-canonical twist n_C/2, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios* live on the rung ladder λ_0 = n_C; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation 0^{++*} = (1,1) sits at the linear conformal energy E = λ_0

+ 2 = 7, degenerate with 2^{++} (2408 MeV; lattice ~2670, within quenched excited-glueball error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-glueball candidate $f_0(1710) \approx 1704$ MeV is consistent with BST's 0^{++} at seat 11 (1720 MeV) — the expected lightest-glueball match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel = 60 = C_2·n_C·rank); with the established χ_top^{1/4} = 180 MeV the WV identity gives f_G(0^{-+}) ≈ 12.6 MeV. Decay-side: mixing (glueball ↔ $q\bar{q}$ overlap on the shared Hardy space), not a "dump."

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg 2^{++} = g/n_C; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^PC tower. It does not close the interacting-vs-free residual (Section 6). That — equivalently, the *formal interacting-RP upgrade* / *proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The so(7) unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: su(3) ⊂ g_2 ⊂ so(7), and so(7) is the compact dual isometry of Q^5 — the same so(7) whose Casimir spectrum is the glueball tower of Section 2; g = 7 is the so(7) vector = g_2 fundamental = $3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain (su(3) not-subset-of so(5,2)) but is a geometric subalgebra of the compact-dual so(7); on the domain side it is the operator algebra on H^2. The discrete-series spectrum of this so(7) has the primaries {N_c, n_C, C_2, g} as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally SU(3) YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with a nonzero, geometry-fixed mass gap (no free dimensionless parameter beyond the ruler; the integer value set by the boundary data $\Delta + \mathbb{Z}_2$-coupling + R on $(S^4 \times S^1)/\mathbb{Z}_2$, K1709).

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values → Bisognano–Wichmann-1975 / Borchers-2000 modular geometry → Rehren-2000 holographic net → Osterwalder–Schrader reconstruction → cluster decomposition.

Rehren's algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_IV^5 (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical SU(3) YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap exists and is nonzero (bounded ⇒ gap) with no free dimensionless parameter beyond the ruler R (Section 2 — the integer value awaits three geometry-fixed opens: the conformal weight Δ, the $\mathbb{Z}_2$ coupling, and R; the bare-S^4 "4" is $\mathbb{Z}_2$-projected-out, K1709); the substrate-derivation of SU(3) (Paper B); the OS *data* (Section 4); the so(7) unification (Section 8). The glueball 1720 MeV at 0.6% is a real, separate forward prediction (built on C_2^int, not λ_1) but out of the room for the value-pin (K1708/K1709). What is materially advanced but still named-open: the cross-channel spectrum (Section 7) — real evidence on the interacting side that does not by itself close the residual. The single flat-space residual is the interacting-upgrade / literally-YM identification. This is a sharpening of the Clay problem — *not* a claim of its solution, and (after K1709) *not* a claim of a derived integer gap value; the gap value is honest open physics, floored at existence.

Explanatory summary (Guard 1 + Guard 2 held). BST *reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain*: the gap $\Delta = C_2 = 6$ is forced by representation theory (Guard 1: a domain operator, not the $\mathbb{R}^4$ Hamiltonian), its pure-glue reading matches the glueball at 0.6% (Guard 2: not the proton), and the flat-$\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- ★ The mass-gap operator on the real boundary (the live task, K1709): define the operator on $(S^4{\times}S^1)/\mathbb{Z}_2$ (not bare S^4), and pin the three geometry-fixed opens — (i) the conformal weight Δ of the lowest gauge excitation, (ii) which surviving $\mathbb{Z}_2$-even (k+m even) mode is lowest, (iii) the radius R (deciding $4/R^2$ vs $4{+}1/R^2$). Glueball out of the room (GATE-KOIDE).
- The one flat-space residual (Section 6/9): the interacting-upgrade / proving the Rehren-OS edge theory is literally SU(3) YM.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear-Schwinger su(3) (algebra closure verified; identification in progress).
- D_A dynamical-gap check (Elie): does the fluctuated Dirac D_A^2 carry a positive gap dynamically, and at what value on the *real* boundary (state the space

first; do not pre-fill "$C_2 = 6$" — that is the projected-out/identity reading)? Report "the D_IV^5 gap," target the glueball, not the proton.

- Falsifier: the cross-channel glueball spectrum is a forward prediction — 0^{++*} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the gap existence-floor and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.7 (K1709): the gap exists (bounded ⇒ gap, no free dimensionless parameter beyond the ruler) — this floor is rock-solid. The *value* is genuine open physics: on the real boundary $(S^4 \times S^1)/\mathbb{Z}_2$ the $\mathbb{Z}_2$ admits only k+m even, so the bare-S^4 "4" (k=1, m=0) is projected out (Cal), leaving R-dependent survivors. The honest value awaits three geometry-fixed opens: (i) the conformal weight Δ, (ii) which $\mathbb{Z}_2$-even mode is lowest, (iii) the radius R (the ruler, deciding $4/R^2$ vs $4+1/R^2$). Energy-vs-Casimir resolved (§570): the ν-independent 4 was the *spatial* S^4 vector; the ν/Δ piece is *time* energy, not mass. The old "6" was the conformal reading = the identity C_2 = n_C+1; "7" = genus — neither is λ_1 on the real boundary. Floor = existence, not ≥ 4. The glueball 1720 is a separate valid prediction (on C_2^int, not λ_1) — kept, out of the room as gap-evidence. Section 0 keeps the three C_2 guards + the glueball exemption. Package intro HELD until Grace ⊕ Elie define the operator on the real boundary and pin Δ + $\mathbb{Z}_2$-coupling + R. NOT a derived value, NOT "spectrum closed," NOT "YM proved." INTERNAL. Nothing pushed; CP existence-only.

Draft v0.7 (K1709). Spine: Cal's $\mathbb{Z}_2$ projection catch (k+m even → bare-S^4 "4" projected out; floor = existence); Grace ⊕ Elie's energy-vs-Casimir resolution (§570: ν-independent 4 = spatial S^4 vector; ν/Δ = time energy, not mass) + the real-boundary operator definition (the live task); Elie's four-formula catch + D_A blind spectrum; Lyra's three-opens framing (Δ + $\mathbb{Z}_2$-coupling + R) + guards; the so(7) unification; Paper B (gauge-group derivation); Keeper K1709 ($\mathbb{Z}_2$ walk-back, floor = existence, glueball over-correction ruled). The value = genuine open physics on $(S^4 \times S^1)/\mathbb{Z}_2$. Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v0.7, K1709: bare-S^4 "4" $\mathbb{Z}_2$-projected-out on the real boundary; floor = EXISTENCE; value = three geometry-fixed opens (Δ, $\mathbb{Z}_2$-coupling, R); glueball separate + out of the room; package intro HELD until the real-boundary operator is defined.